MA Driver's Manual Updates — Project Brief, PRD & Tradeoffs Analysis

Status: Working prototype (internal)

Audience: RMV leadership and IT/web stakeholders

Purpose: Decision-support for finishing and extending the tool

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Executive brief

One page for decision-makers. The full product requirements and cost analysis follow as backup (Sections 1–8).

> **See it live: https://szgrune.github.io/rmv-manual-updates/web/** — the working app is the fastest way to grasp this.

The problem

Massachusetts has on the order of **5 million licensed drivers**. Most earned their license as teenagers and haven't opened the driver's manual since. Yet the rules keep changing — hands-free phone requirements, new protections for cyclists and other vulnerable road users, cannabis and impaired-driving guidance — and today there is **no easy way for a driver to find out what changed since they learned to drive.** The official manual is ~140 pages, republished as a whole, with the changes buried inside it.

The solution

A simple public website answers one question: *"What has changed in the driver's manual since I got my license?"* A driver enters the year they were licensed and immediately sees a clear, organized summary of what's new, updated, or removed — with **every item linked straight to the exact page of the official RMV manual.** It is an access-and-orientation layer on top of the RMV's own content; the manual remains the source of truth.

Why MassDOT, why now

- **It serves people the RMV already serves** — whether court-mandated driver-education participants or newly relocated drivers — and routes them to official content faster.
- **It's a low-risk, high-visibility example of responsible AI in state government:** built quickly with AI assistance, grounded in the official document, with a citation on every claim, and designed intentionally to serve public interest.
- **A working prototype already exists.** This prototype was built in the first week of the fellowship — a demonstration of what AI-assisted development makes possible. Production-quality work — security, accessibility, self-service tools for RMV staff, additional languages — is addressed in the subsequent sections of this brief.

What it costs

- **To operate: under $100 a year.** The only recurring cost is the AI analysis that runs when a new manual is released — a few dollars per run.
- **To finish: one intern-summer**, with light in-house IT/web follow-up. No new budget of consequence.
- **IT footprint: negligible.** The public site is static — no new server, database, or software licenses required (it relies only on the existing MassDOT OpenAI license to stay operational).

What we're asking

Help move this from prototype to public service by:

1. **Endorsing it to the RMV** and connecting us with the right digital/content owners;
2. **Approve completion** of the remaining work through this summer's internship plus light in-house follow-up.

Questions of accuracy, equity, and legal exposure are real and have been thought through; they are addressed candidly in Section 3 and the appendix.

Full PRD & tradeoffs analysis follows.

> **Note on figures.** Time and cost figures in this document are planning estimates, expressed as ranges. Labor rates reflect the assumptions confirmed for this project (see Cost & effort model). LLM/API pricing is approximate and should be verified against current vendor rates before budgeting.

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1. Executive summary

The **Driver's Manual Updates by Year** app answers a single, common question for Massachusetts drivers: *"What has changed in the driver's manual since I got my license?"* A user enters the year they were licensed, and the app shows a clear, organized summary of what is new, updated, expanded, or removed in the current manual compared to the edition closest to that year — with each quoted change linked back to the exact page of the official RMV PDF.

How it works (in brief). A small Python pipeline ingests each manual PDF, uses a large language model (LLM) to compare a newer edition against an older one and describe the substantive changes, and exports the result to a lightweight website. Every quoted change carries a **page-level citation** that deep-links to the official manual PDF, so the authoritative source is always one click away.

How it was built. The prototype was developed rapidly by co-developing with agentic AI coding tools (Claude Code / Codex). This is the same workflow proposed for finishing the product, and it is the primary reason the build estimates below are well under traditional software timelines.

Primary users. Newly relocated and returning drivers, driver-education students and instructors, citizens taking court mandated driving education, mature drivers or people who have had their licenses for a long time, and RMV content owners — see user typologies.

Current status. The English app is functional with four editions (2007, 2017, 2023, 2026), search, change-type filters, share/print-to-PDF, page citations, and a curated content pass that removes low-value "statistics changed" noise. What remains is to make it **operationally self-service** (so RMV staff can add a new manual each year without a developer), to make the **images and text content easily editable**, and to **broaden coverage** (more editions, accessibility, and languages).

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2. Project brief

Overview

The tool turns a tedious, error-prone task — manually comparing a ~140-page legal document against a prior edition — into a guided, citable summary. It is designed to be **informational**, not authoritative: the official RMV manual remains the source of truth, and the app's job is to *route users to the relevant official content faster*, with citations on every claim.

The system has two halves:

- **Build pipeline (back office).** `extract_pdfs.py` → `analyze_changes.py` → `export_json.py` (~1,900 lines of Python) over a local SQLite database, plus a permanent **manual-override layer** so any human correction survives future re-runs, and a **citation matcher** that pins each quoted change to its physical PDF page (currently matching 296 of 300 quotes, ~98%).
- **Public website (front office).** A fast, dependency-light static site (`index.html`, `app.js`, `styles.css`) — no server or database required to serve the public experience.

Key user typologies

User	What they need	How the app serves them
Returning driver (licensed years ago, re-engaging)	A quick read on what's changed during their time away	Enter your license year → "Changes Since {year}"
Newly relocated driver	Orientation to MA-specific rules	Closest-edition comparison + Rules of the Road surfaced first
Driver-education student / instructor	Teach to the *current* manual, flag what's new	Filterable, search-able change list with citations
Court-mandated driving-education participant	Confirm the current rules they're required to learn	Authoritative, citation-backed summary of what's current
Mature / long-licensed driver	Catch up on rules that changed over many years	Earliest-edition comparison surfaces the most accumulated change
RMV content owner / staff	Add each new manual; fix misplaced images and text	Self-service admin + image/content CMS (planned)

How it works (pipeline detail)

1. **Extract** — each manual PDF is parsed into structured sections, text, page numbers, and images.

2. **Analyze** — the newer manual is chunked and compared, chunk-by-chunk, against the *complete* text of the older manual; the LLM returns a structured list of substantive changes (new / updated / expanded / removed).
3. **Curate** — human corrections are captured permanently in an overrides layer; low-value items (e.g., refreshed statistics) are pruned.
4. **Cite & export** — every quote is matched to its PDF page; results are written to a JSON file that the website reads.

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3. Concerns & counterarguments

This section anticipates the questions a careful reviewer should raise, with the mitigation for each.

Concern	Assessment & mitigation
It may not be exhaustive — could the AI miss a change?	Correct — this is an *assistive summary*, not a substitute for the manual. Mitigations: (1) every claim links to the official PDF page; (2) a permanent human-override layer lets staff add, edit, or remove items; (3) a prominent "informational only — consult the official manual" disclaimer. The tool's value is *triage and access*.
AI accuracy / hallucination	The analysis quotes the manual directly and pins each quote to a page; reviewers can verify in one click. Quotes that cannot be matched to a page are shown without a (broken) link rather than with a guessed one. A human QA pass is built into the per-edition workflow.
English/Spanish-only — an equity issue?	Framed neutrally: language access is an accessibility and service-delivery question. The RMV already publishes the manual in multiple languages; the roadmap mirrors that. The tool is built so additional languages are incremental, without requiring rewrites (see Phase 2). Prioritization is a policy decision for RMV, not a technical constraint.
English/Spanish-only — a political issue?	The recommended posture is to follow the RMV's existing published-language set and official translations, so the tool inherits the agency's language policy rather than setting its own. Where no official translation exists, machine translation of legal content is **not** recommended without human review.
Content freshness	Updates are tied to manual releases (infrequent). The self-service admin (Phase 1) ensures a new edition can be added the same week it's published, without developer involvement.
Legal exposure	Mitigated by an explicit "informational, not legal advice; the official RMV manual governs" disclaimer and visible source citations.
Privacy	The public app collects no personal information and requires no login; a year typed into a box is not stored or transmitted with identity.
Accessibility	A dedicated Phase 1 work item brings the app to WCAG 2.1 AA (keyboard, screen-reader, contrast), appropriate for a public-sector service.

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4. Cost & effort model

Labor assumptions (confirmed):

- **Intern:** $22/hr → **~$176 per 8-hour day**. Availability this summer: ~8 weeks ≈ **320 hours ≈ ~40 developer-days**.
- **In-house IT/web follow-on:** ~$40/hr → **~$320 per day**.
- **1 "dev-day" = 8 focused hours.** Ranges are low–high; "expected" is the planning midpoint.

AI acceleration. All effort estimates assume the developer co-develops with agentic AI (Claude Code / Codex), consistent with how the prototype was built. A **traditional, no-AI baseline (~2–3×)** is shown alongside to make the savings explicit.

The only required non-labor cost: annual LLM tokens. When a new manual is released, the comparison is re-run. Because each chunk of the new manual is compared against the *full* text of each older edition, a single comparison pair costs roughly **0.7M–1.0M input tokens**. Today's three pairs total ≈ **2.5M input + ~0.4M output tokens** per full re-run.

Model class (approx. pricing)	Cost per full annual re-run (today, 3 pairs)	At larger scale (~8 pairs)
Premium (Opus-class, ~$15/M in, ~$75/M out)	**~$65–75**	~$150–200
Mid (Sonnet-class, ~$3/M in, ~$15/M out)	**~$15**	~$35–45
Efficient (Haiku-class, ~$1/M in, ~$5/M out)	**~$5**	~$10–15

The prototype currently defaults to the premium model. For production, a mid-tier model is recommended for this extraction task, cutting cost ~5×. **Net: the recurring operating cost is on the order of $15–75/year now, and likely under a few hundred dollars/year even at full multi-edition scale** — a rounding error compared to labor. A small one-time token budget during development (re-running the analysis while tuning) is similarly minor (~$15–75 total).

Hosting is negligible: the public site is static and can run on existing state web infrastructure or a static host; the admin backend (Phase 1) needs only a small server/runtime.

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5. Phase 1 — Finalize the existing product

These items make the tool **self-service, accurate, accessible, and bilingual**.

5.1 Self-service admin backend (upload a manual → run → publish)

Today, adding a manual requires a developer to run command-line scripts. This builds a simple, authenticated web admin where RMV staff **upload a new manual PDF, kick off the extract→analyze→export pipeline as a background job, watch progress, review the results, and publish** — wrapping the existing, proven scripts rather than rewriting them.

5.2 Image CMS (easy, visual image editing)

Extracted images frequently land in the wrong update or position. The override layer already supports image edits in data; this adds a **friendly visual editor** to drag images to the correct change, reorder, add, or remove them, with live preview — no JSON editing.

5.3 Include all editions 2007–2026

Ingest and QA every available edition in the range so any driver lands on a genuinely close comparison. *Estimate is per-edition.*

5.4 Accessibility improvements (WCAG 2.1 AA)

Keyboard navigation, ARIA roles/landmarks, focus management, color-contrast, and screen-reader testing — appropriate for a public-sector service.

5.5 More accurate / streamlined content

Refine the analysis prompts and add a light relevance-curation step so results emphasize the **most decision-relevant** changes (rules and laws) and continue to suppress low-value noise (e.g., refreshed statistics), with a human review pass.

5.6 Spanish version

Ingest the official RMV Spanish manual(s), run the (reusable) pipeline, and add UI internationalization with a language toggle and Spanish-language QA. **This item also builds the one-time i18n framework** that makes every later language cheap.

Phase 1 estimates

Work item	Effort w/ AI (days)	Effort no-AI (days)
5.1 Admin backend	10–18	25–45
5.2 Image CMS	8–15	20–38
5.3 All editions 2007–2026	6–12	12–24
5.4 Accessibility (WCAG 2.1 AA)	5–9	10–18
5.5 Streamlined content	5–10	10–20
5.6 Spanish version (incl. i18n)	8–14	20–35
Phase 1 total	**42–78** (≈59)	**97–180** (≈138)

Reading the table. With AI, Phase 1 is roughly **42–78 developer-days** (≈59). The traditional, no-AI baseline is **~97–180 days**, so the AI workflow removes on the order of **55–60% of the effort**.

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6. Phase 2 — Potential add-ons (optional)

These extend reach but are not required for a credible launch. They become inexpensive *because* Phase 1 builds the admin, image CMS, and i18n framework.

6.1 Additional languages

Mandarin, Arabic, Korean, Haitian Creole, Portuguese, French, Russian, etc. Once the Spanish work establishes the i18n framework, each language is mostly: ingest the official translated manual, run the pipeline, translate UI strings, and QA.

- **Per language: ~3–6 days. Arabic adds right-to-left (RTL) layout: ~5–9 days.**
- **Gating caveat:** this assumes an **official translated manual exists** to ingest. Where one does not, machine-translating official legal content is **not recommended** without human review — that becomes a larger, policy-gated effort.

6.2 Digitize pre-2007 manuals (OCR)

Older editions may exist only as scanned images, requiring OCR plus heavier extraction QA before they can enter the pipeline.

- **Per manual: ~3–6 days**, OCR-quality dependent.
- **Caveat:** source availability and scan quality drive the range; old scans need more human QA than born-digital PDFs.

Phase 2 estimates

Work item	Effort w/ AI (days)	Effort no-AI (days)
6.1 Languages (per language)	3–6	7–14
6.1 Arabic (RTL premium)	5–9	11–20
6.1 All 7 listed languages (sources permitting)	30–40	60–85
6.2 Pre-2007 manuals (per manual)	3–6	6–12
6.2 Pre-2007 (≈2–3 older editions)	8–18	16–36

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7. Roll-up & recommendation

Scope	Effort w/ AI (days)	Effort no-AI (days)
Phase 1 (committed)	42–78	97–180
Phase 2 — all 7 languages	+30–40	+60–85
Phase 2 — pre-2007 (≈2–3)	+8–18	+16–36

Recurring (annual): ~$15–75/yr in LLM tokens (mid-tier model) — the only required non-labor operating cost.

What fits the internship. The intern's ~40 available days covers a large share of Phase 1 but not all of it. A practical split:

1. **Intern (this summer, ~40 days):** Admin backend (5.1), Image CMS (5.2), and the all-editions ingest (5.3) — the operational core that makes the tool self-sustaining.
2. **In-house IT/web follow-on:** finish accessibility (5.4), streamlined content (5.5), and the Spanish version + i18n framework (5.6).
3. **Later / as prioritized:** Phase 2 languages and pre-2007 digitization, each cheap once the framework exists.

Bottom line. Finishing the initial scope is roughly a **one-summer-plus effort (~42–78 developer-days)**, with a recurring cost measured in **tens of dollars per year**. The AI-co-development workflow roughly halves the labor versus a traditional build, and the architecture is deliberately structured so that breadth (more editions, more languages) is additive rather than a rebuild.

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8. Appendix

Assumptions

- 1 dev-day = 8 focused hours. Intern $22/hr ($176/day); in-house $40/hr ($320/day).
- AI-accelerated effort assumes co-development with Claude Code / Codex; no-AI baseline ≈ 2–3×.
- "All editions 2007–2026" assumes that PDFs are obtainable from the RMV.
- Additional-language and pre-2007 estimates assume an **official source document exists**; absence of an official source materially changes scope and policy risk.

Pricing / model note

LLM token prices are approximate and tiered by model (premium / mid / efficient); **verify current vendor rates before budgeting.** The prototype defaults to a premium model; a mid-tier model is recommended for production extraction, reducing recurring cost ~5×.

Grounded reference figures (current build)

- Editions: 2007 (159 pp), 2017 (166 pp), 2023 (120 pp), 2026 (140 pp).
- Per comparison pair: ~0.7M–1.0M input tokens; current 3 pairs ≈ 2.5M in + ~0.4M out/run.
- Citation match rate: 296/300 quotes (~98%).
- Codebase: ~1,900 lines of pipeline Python; static frontend; SQLite + JSON output.

Glossary

- **Edition / pair** — a manual year; a "pair" is one older→newer comparison.
- **Override layer** — stored human corrections that always win over AI output and survive re-runs.
- **Citation** — a quote's deep link to the exact page of the official RMV PDF.
- **i18n** — internationalization; the one-time framework enabling multiple languages.
- **WCAG 2.1 AA** — the accessibility standard targeted for the public service.